Haohao Sun

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Education

- **Nanjing University** — Nanjing, China
  Ph.D. - Meteorology — *Sep 2020 - June 2026*
  Advisor: Prof. Lili Lei
- **Nanjing University** — Nanjing, China
  B.S. - Atmospheric Science — *Sep 2016 - Jun 2020*

Research Interests

My research focuses on paleoclimate data assimilation (PDA), climate field reconstruction, and the integration of artificial intelligence (AI) with data assimilation. I am currently working on climate reconstructions of the Last Millennium to improve our understanding of climate variability and dynamics across multiple timescales.

1. Offline PDA: Paleoclimate data assimilation is most commonly implemented in an *offline* framework, where geological proxy records are combined with pre-computed climate model simulations. I developed an Integrated Hybrid Ensemble Kalman Filter (IHEnKF) that combines small-scale flow-dependent and large-scale climatological information, providing a computationally efficient framework while maintaining high reconstruction skill.

2. Online PDA: By replacing the numerical climate model with an AI-based surrogate model, I developed an *online* PDA framework based on IHEnKF that propagates proxy information forward in time. Furthermore, I proposed an Integrated Hybrid Ensemble Kalman Smoother (IHEnKS), which assimilates future proxies to further improve the reconstruction of past climate states.

Selected Publications

- **Sun, H.**, L. Lei, Z. Tan, L. Ning, and Z. Liu, 2026 "An Ensemble Kalman Smoother for Online Paleoclimate Data Assimilation", *J. Adv. Model. Earth Syst.*. (Minor Revision).
- Lei L., **Sun, H.**, Z. Tan, L. Ning, and Z. Liu, 2026 "Strongly Coupled Data Assimilation for Paleoclimate Reconstruction Using Deep Learning-Based Models.", *J. Adv. Model. Earth Syst.*, **18**, e2025MS005686, https://doi.org/10.1029/2025MS005686.
- **Sun, H.**, L. Lei, Z. Liu, L. Ning, and Z. Tan, 2025 "An Online Paleoclimate Data Assimilation with a Deep Learning-based Network", *J. Adv. Model. Earth Syst.*, **17**, e2024MS004675, https://doi.org/10.1029/2024MS004675.
- **Sun, H.**, L. Lei, Z. Liu, L. Ning, and Z. Tan, 2024 "A Hybrid Gain Analog Offline EnKF for Paleoclimate Data Assimilation", *J. Adv. Model. Earth Syst.*, **16**, e2022MS003414, https://doi.org/10.1029/2022MS003414.
- **Sun, H.**, L. Lei, Z. Liu, L. Ning, and Z. Tan, 2022 "An Analog Offline EnKF for Paleoclimate Data Assimilation", *J. Adv. Model. Earth Syst.*, **14**, e2021MS002674, https://doi.org/10.1029/2021MS002674.
- Z. Wang, **Sun, H.(co-first-author)**, L. Lei, Z. Tan, and Y. Zhang, 2023 "The Importance of Data Assimilation Components for Initial Conditions and Subsequent Error Growth", *Sci China Earth Sci.*, **67**, 105-116, https://doi.org/10.1007/s11430-023-1229-7.

Project Management

- National Natural Science Foundation of China(Basic Research Program for Young Students), 2024-2026 "Ensemble-based Multi-scale Data Assimilation for Last Millennium Climate Reconstruction", **leader**
- National Key Research and Development Program of China, 2023-2028 "Data Assimilation Theory and Techniques for Holocene Climate Change in the Tropical Western Pacific", participation

Teaching

- **Teaching Assistant** — Nanjing University
  Graduate course: Data Assimilation — *Spring 2022*
- **Teaching Assistant** — Nanjing University
  Undergraduate course: Dynamic Meteorology — *Fall 2021*

Awards

- National Scholarship - 2024, CNY 30000
- Presidential Scholarship - 2020, CNY 20000
- Environmental Planning Scholarship - 2022, CNY 5000

Programming

MATLAB/Python(numpy, scipy, xarray, Tensorflow, Pytorch)/Fortran/Linux/C